USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 21 LAB - Antenna Pattern
Lab Packet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 3.6: I can distinguish between array factor and true antenna pattern and account for element pattern effects.

This packet is the L21 Lab, in three parts, and you turn it in once, complete, before Lesson 22. Do **Part 1** on your own before the lab period: what you should see at the bench, how to pace a sweep by hand, and what counts as agreement. You need it at the bench. **Part 2** is your own measurements, from your own kit, recorded in class. Do **Part 3** on your own after the lab: compare the methods lobe by lobe, compare the sweep by hand with the pass on the stand, and work a far-field problem.

Grading. One submission, counted toward your engagement grade and graded as a complete, documented attempt.

The array is the PHASER's: $N = 8$ elements, E1 to E8, on a $d = 14$ mm pitch, so the aperture is $D = Nd = 112$ mm, one spacing for each element (Lesson 18). Facing the patches, E1 is on the right and E8 on the left, and $+\theta$ is on the E8 side. Use $c = 3.00 \times 10^8$ m/s, and report every lobe level relative to its own trace's main-lobe peak, in dBc, unless a question asks otherwise.

Part 1: Pre-Lab (individual, before the lab period)

Part 1 works out what you should see at the bench, so you can tell a good trace from a bad one while you are still at the kit.

1. **Frequency and distance.** Use the HB100 frequency your kit reported with **Find HB100** in the L18 Lab. If you did not record one, use the nominal 10.525 GHz and say so. The lesson's figures use 10.3 GHz, so your angles will differ from them by a few tenths of a degree.

   (a) Record the frequency you are using, then find the wavelength, the electrical spacing d/λ, and kd in degrees.

   f = _____

   d/λ = _____

   kd = _____

   (b) Find the far-field distance of the array at your frequency, and say whether the lab's 1.2 m clears it.

   r = _____

2. Reading dBc off a trace. The kit reports every level in dBFS, relative to the receiver's full scale. A trace gives peak readings of -18.4 dBFS for the main lobe, -31.2 dBFS for the left first sidelobe, and -29.9 dBFS for the right first sidelobe.

(a) Convert both sidelobe readings to dBc, the level relative to the trace's own main-lobe peak.

L_{left} = ____

L_{right} = ____

(b) Raising **Rx Gain** by 6 dB and repeating the measurement changes the three dBFS readings. State what happens to them and to the two dBc values, and explain why that lets you compare a beam sweep with an azimuth sweep taken minutes apart.

3. The beam at broadside. Table B is a beam sweep with the HB100 at boresight; Tables C and D hold the beam at 0° and move the source in azimuth. Both trace the array factor

$$|AF| = \left| \frac{\sin(N\psi/2)}{N \sin(\psi/2)} \right|$$
$$\psi = kd\,(\sin\theta_{\text{src}} - \sin\theta_0)$$

so predict one set of features for all three.

(a) Find the angles of the first three nulls on each side.

θ_1 =

θ_2 =

θ_3 =

(b) The array factor puts the first sidelobes near $\pm 21.5^\circ$, 12.8 dB below the peak. The azimuth sweep also carries the source across each patch's own pattern, which falls toward $\pm 90^\circ$. Should that make the azimuth sweep's first sidelobes differ from the beam sweep's by more than the 2 dB a room can move a dBc value? Give the reason.

(c) Part 2 records the level at 10° marks of the stand. Name the marks that fall within a few degrees of a null, and explain in one or two sentences why the marks alone cannot give you the sidelobe levels.

4. The beam at 30°. Table E holds the beam at $\theta_0 = 30^\circ$ and turns the kit.

(a) Find the first null on each side of the steered beam.

inner = ____

outer = ____

(b) The array factor puts both first sidelobes 12.8 dB below the peak, the inner one near 8° and the outer one near 60°, and a beam sweep would show exactly that. Turning the kit also carries the source across the patch pattern, which the course models as $\cos\theta$ in power. Table E reads in dBc of its own peak, near 30°. Which first sidelobe of Table E should read lower than the beam sweep's, and by about how much?

lower = ____

by = ____

(c) Your room can make the two first sidelobes of a broadside trace differ by a few decibels on its own. What must you compare the Table E difference with before you credit it to the patch pattern?

(d) Part 2 also takes one reading with the kit turned to 180°, the source behind the board. Predict that reading, and give the reason.

5. Pacing the sweep by hand. In Table C you carry the HB100 from -90° to $+90^\circ$ in about 22 s while Signal vs Time plots the level against time.

(a) At a steady pace, how many degrees per second is that, and how long does the source spend inside the 13.0° half-power width of the main lobe?

rate = _____

t = _____

(b) A first sidelobe spans the gap between the first and second nulls. How long does the source spend in it, and what is the slowest update rate, in updates per second, that puts at least five readings on it?

t = _____

rate = _____

6. **What counts as agreement.** Part 2 sweeps the pattern in azimuth twice with the beam held fixed. In the **sweep by hand** you carry the HB100 around a taped arc of radius 1.2 m. In the **pass on the stand** the HB100 stays on its tripod 1.2 m out, and you turn the kit about the center of the array, reading the level at the stand's marks.

   (a) During a sweep by hand your radius drifts by 10 cm either way. Received power goes as $1/r^2$. Find the change in the reading for a radius of 1.1 m and of 1.3 m.

   1.1 m: ____

   1.3 m: ____

   (b) If you convert the time axis to angle assuming a steady pace, and your hand moves up to 30% faster or slower than average while crossing the main lobe, give the range of half-power beamwidths the conversion reports for a true 13.0° beam.

   range = ____

   (c) On the stand, suppose the kit turns about a point 20 mm from the center of the array. Find the largest angle error and the largest change in level this offset can produce at 1.2 m.

   $\Delta\theta$ = ____

   ΔL = ____

   (d) A reflection from the room adds to the direct path in both methods. Find the relative amplitude ρ of a single reflected path that moves the reading by $+1$ dB when the two add in phase, and express ρ in dB.

(e) The main-lobe peak and a sidelobe peak each carry about ± 1 dB from reflections. Give the worst-case uncertainty on the sidelobe level in dBc, and the range a true -13 dBc sidelobe could read. This is the window you will use to decide whether two traces agree.

uncertainty = ____

range = ____

Part 2: Measurements (your kit, in class)

Follow the procedure on the L21 Lab page. Before you record anything, set **Rx Gain** so that the main-lobe peak, with the source at boresight and the beam at 0°, reads -10 ± 3 dBFS on the FFT, and leave it there for the whole lab. Keep the HB100 at patch-row height, 1.2 m from the array. Record each reading as you take it.

Signal Freq from **Find HB100**: _____ Rx Gain: _____ Source distance: _____ Updates/s: _____

7. **Table A: check the angle axis.** With the beam sweeping as in the L18 Lab, turn the kit so the source sits at each angle below, and record the **Angle** readout, set to **Peak**. Turning the kit's face toward E1, as seen from the source, puts the source on the $+$ side, and the stand's scale reads the opposite sign: -30 on the stand puts the source at $+30^\circ$.

Source angle (stand reads)	-30° ($+30$)	0° (0)	$+30^\circ$ (-30)
Angle (Peak)			

8. **Table B: the beam sweep.** Kit back at 0°, beam sweeping. Read the trace.

Peak (dBFS)	**HPBW**	**Left sidelobe (dBc)**	**Right sidelobe (dBc)**

9. **Table C: the azimuth sweep by hand.** Beam held at 0°. Carry the HB100 from -90° to $+90^\circ$ on the taped 1.2 m arc, aimed at the array, three times. Record each sweep's main-lobe peak and both first sidelobes.

	Sweep time (s)	**Peak (dBFS)**	**Left SL (dBc)**	**Right SL (dBc)**
Sweep 1				
Sweep 2				
Sweep 3				
Spread (max − min)				

10. Table D: the pass on the stand, beam held at $0°$. HB100 on its tripod at boresight, 1.2 m out, beam sweeping. One pass fills Tables D and E: each sweep sample is the beam held at that steer angle, so the trace at $0°$ steer is this table and the sample nearest $30°$ is Table E. Turn the kit to each mark of the stand and record the trace at $0°$ relative to its value with the kit at $0°$, in dB, at the source angle: the stand reading with its sign flipped, as in Table A. Write "floor" where the reading stops changing as you turn.

θ_{src}	$-90°$	$-80°$	$-70°$	$-60°$	$-50°$	$-40°$	$-30°$	$-20°$	$-10°$	$0°$
Level										

θ_{src}	$+10°$	$+20°$	$+30°$	$+40°$	$+50°$	$+60°$	$+70°$	$+80°$	$+90°$	
Level										

Now turn the kit slowly through each lobe and null below, and record the angle and level of its extreme, the highest reading for a lobe and the lowest for a null.

	Source angle	**Level**
Main lobe		
Left first null		
Right first null		
Left first sidelobe		
Right first sidelobe		

11. Table E: the pass on the stand, beam held at 30°. The same pass, read at the sweep sample nearest 30°. Record each level relative to the *Table D* reference.

θ_{src}	-90°	-80°	-70°	-60°	-50°	-40°	-30°	-20°	-10°	0°
Level										

θ_{src}	$+10^\circ$	$+20^\circ$	$+30^\circ$	$+40^\circ$	$+50^\circ$	$+60^\circ$	$+70^\circ$	$+80^\circ$	$+90^\circ$	180°
Level										

	Source angle	**Level**
Main lobe		
Inner first null		
Outer first null		
Inner first sidelobe		
Outer first sidelobe		

Part 3: Writeup (individual, after the lab)

12. Compare the methods lobe by lobe. Answer each part in three or four sentences, using your numbers from Part 2, your predictions from Part 1, and the ± 2 dB window of Question 6.

(a) At broadside, compare the first sidelobes of the beam sweep (Table B), the sweeps by hand (Table C), and the pass on the stand (Table D). Do the three methods agree?

(b) With the beam at 30°, compare the inner and outer first sidelobes of Table E with your answer to Question 4(b). Is the asymmetry in the predicted direction, and is it larger than the broadside asymmetry of Table D?

(c) Report your 180° reading and explain it. Then explain why the beam sweep of Table B cannot show the patch pattern, however carefully it is taken.

13. The sweep by hand against the pass on the stand. Use Table C and Tables D and E.

(a) Compare the spread of your three sweeps by hand with the repeatability you saw on the stand. Could a single sweep by hand show the 2 dB outer-sidelobe difference at 30°? Give the reason.

(b) A sweep by hand runs from $t = 0.0$ s at -90° to $t = 22.0$ s at $+90^\circ$, with the beam held at 0°, and the main-lobe peak lands at $t = 9.2$ s. Convert that time to an angle assuming a steady pace, and state the error.

θ =

error =

(c) Explain why an uneven pace moves the lobe positions on a sweep by hand but not the lobe levels, and name what an antenna range uses to record the angle of every sample.

(d) A customer wants written proof that this array's sidelobes are below -25 dBc. State whether either of today's measurements can support that claim, and give the two reasons that decide it.

14. How far out is far enough. A Ku-band panel with a 0.200 m aperture is to be measured at 15.0 GHz.

(a) Find its far-field distance.

r =

(b) The panel is measured on today's 1.2 m range anyway. The extra path from the source to the aperture edge, relative to its center, is about $D^2/8r$. Find the edge phase error and compare it with the 22.5° the far-field criterion allows.

$\Delta\phi$ =

(c) Describe what that phase error does to the measured pattern, and say which pattern feature it corrupts worst.

Documentation: _____